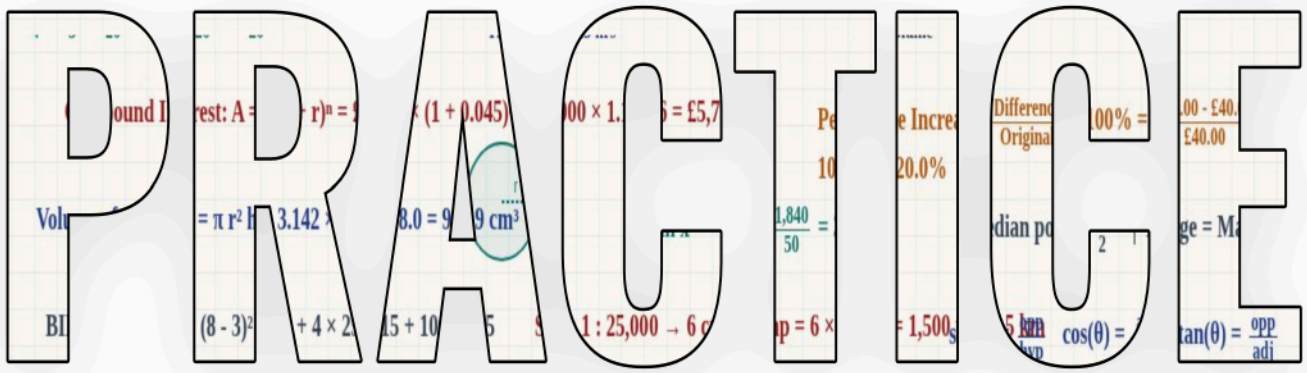
**MASTER FUNCTIONAL SKILLS**

FUNCTIONAL SKILLS MATHEMATICS

COMPANION PRACTICE WORKBOOK • STEPPED DRILLS

LEVEL 2




# PRACTICE COMPANION

120+ stepped exam drills & full method mark workspaces

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MATHS LEVEL 2

# Functional Skills Mathematics

Level 2 Companion Workbook • 5mm Grid Candidate Working Frames & Full Mocks

| This book belongs to: | Exam date: | Target level: |
|---|---|---|
| _____ | _____ | _____ |

## What is inside


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## How to use this book

**Read**
Read the rule. Study the picture.

**Try**
Write your answers on the lines.

**Check**
Mark it with the answers at the back.

**Remember**
Learn the Golden Rule by heart.

DRILL 1 OF 34 • LEVEL 2 STANDARDS

# 1 Worksheet 1: Negative Numbers & Overdrafts

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Rapid Warm-Up Drills:** Evaluate without a calculator: **(4 marks)**

a) -14 + (-9) = _____ b) 12 - (-18) = _____

c) (-6) × (-7) = _____ d) (-72) ÷ (+8) = _____

**Show your working**

**Answer:** _____

**2** **Task 2: Workplace Context — Commercial Overdraft:** A transport firm has a bank overdraft balance of **-£2,340** on Monday morning. During the week: client payment +£5,800, diesel fuel -£1,950, insurance direct debit -£780, and maintenance -£1,150. Calculate the closing balance. **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 1: Negative Numbers & Overdrafts • Candidate Workspace

✎ Show all working for Worksheet 1 tasks below. Check your final answers.

DRILL 2 OF 34 • LEVEL 2 STANDARDS

# 2 Worksheet 2: Column Multiplication & Decimals

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Column Multiplication Warm-Up:** Calculate without using a calculator. Show your column or grid workings clearly: **(4 marks)**

a) 348 × 27 b) 516 × 38

**Show your working**

**Answer:** _____

**2** **Task 2: Bus Stop Division with Decimals:** Divide **918 by 24** using short or long division. Give your answer as an exact decimal. **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 2: Column Multiplication & Decimals • Candidate Workspace

✎ Show all working for Worksheet 2 tasks below. Check your final answers.

DRILL 3 OF 34 • LEVEL 2 STANDARDS

# 3 Worksheet 3: Fraction Operations & Timber

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Fraction Operations Warm-Up:** Evaluate giving answers in simplest form: a) 5/12 + 3/8 b) 2 1/4 - 1 2/3. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Joinery Construction Problem:** A joiner cuts two battens: 1 3/4 m and 2 3/8 m from a 5 m timber plank. What length of plank remains? **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 3: Fraction Operations & Timber • Candidate Workspace

✎ Show all working for Worksheet 3 tasks below. Check your final answers.

DRILL 4 OF 34 • LEVEL 2 STANDARDS

# 4 Worksheet 4: Fractions Multiply/Divide & Bakery

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Multiplication & Division Drills:** Evaluate giving answers in simplest form: a) 4/9 × 3/8 b) 3 1/2 ÷ 7/10. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Commercial Bakery Batching:** A baker uses 2 1/4 kg of flour per cake batch. He has 18 kg of flour in stock. How many batches can he bake? **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 4: Fractions Multiply/Divide & Bakery • Candidate Workspace

✎ Show all working for Worksheet 4 tasks below. Check your final answers.

DRILL 5 OF 34 • LEVEL 2 STANDARDS

5

# Worksheet 5: Decimals Arithmetic & Garage Invoice

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Decimal Arithmetic Warm-Up:** Calculate: a) 24.85 + 9.382 **(4 marks)**
b) 100 - 37.64 c) 4.6 × 0.35 d) 15.75 ÷ 0.5.

**Show your working**

**Answer:** _____

**2** **Task 2: Mechanics Parts Invoice:** An auto garage invoice lists: 4 spark plugs at £14.25 each, brake pads at £68.50, and 2.5 hours labour at £42.00/hr. Find the pre-VAT total. **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 5: Decimals Arithmetic & Garage Invoice • Candidate Workspace

✎ Show all working for Worksheet 5 tasks below. Check your final answers.

DRILL 6 OF 34 • LEVEL 2 STANDARDS

6

# Worksheet 6: Percentages & Clearance Sale

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Multiplier Quick Fire:** State the single multiplier for: a) Increase by 17.5% b) Decrease by 12% c) Increase by 4.5% d) Find 8.25%. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Retail Clearance Discount:** A television originally priced at £650.00 is discounted by 22%. Calculate the new sale price. **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 6: Percentages & Clearance Sale • Candidate Workspace

✎ Show all working for Worksheet 6 tasks below. Check your final answers.

DRILL 7 OF 34 • LEVEL 2 STANDARDS

# 7 Worksheet 7: Reverse Percentages & VAT

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Reverse Multiplier Warm-Up:** Find the original value: a) Value after 20% increase is £144 b) Value after 15% decrease is £204. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Pre-VAT Trade Price:** A tradesperson buys a cement mixer for £348.00 including 20% VAT. How much did the mixer cost before VAT was added? **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 7: Reverse Percentages & VAT • Candidate Workspace

✎ Show all working for Worksheet 7 tasks below. Check your final answers.

DRILL 8 OF 34 • LEVEL 2 STANDARDS

# 8 Worksheet 8: Ratio Sharing & Business Profit

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Ratio Warm-Up Drills:** a) Share £560 in ratio 5 : 2. b) Express the ratio 15 cm : 1.2 m in simplest form 1 : n. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Partnership Profits:** Two directors share company profit in ratio 7 : 4. Director A receives £21,000 more than Director B. What was total profit? **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 8: Ratio Sharing & Business Profit • Candidate Workspace

✎ Show all working for Worksheet 8 tasks below. Check your final answers.

DRILL 9 OF 34 • LEVEL 2 STANDARDS

# 9 Worksheet 9: Proportion & Unit Value

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Inverse Proportion Warm-Up:** a) 4 builders take 6 days to lay paving. How long for 3 builders at same pace? b) A car at 60 mph takes 4 hours. How long at 48 mph? **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Unit Pricing Best Buy:** Paint Brand X: 5 litres for £28.50. Paint Brand Y: 7.5 litres for £41.25. Which offers better value per litre? **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 9: Proportion & Unit Value • Candidate Workspace

✎ Show all working for Worksheet 9 tasks below. Check your final answers.

DRILL 10 OF 34 • LEVEL 2 STANDARDS

10

# Worksheet 10: Compound Interest & Van Depreciation

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Compound Multiplier Drills:** Calculate: a) £4,000 at 3% compound interest for 2 years. b) £12,000 depreciating at 10% for 3 years. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Fleet Van Depreciation:** A delivery van cost £24,000 brand new. It depreciates by 15% in year 1 and 12% in year 2. What is its value after 2 years? **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 10: Compound Interest & Van Depreciation • Candidate Workspace

✎ Show all working for Worksheet 10 tasks below. Check your final answers.

DRILL 11 OF 34 • LEVEL 2 STANDARDS

# 11 Worksheet 11: BIDMAS & Braking Formula

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: BIDMAS & Powers Warm-Up:** Evaluate: a) $18 - 4 \times 3 + 5$ **(4 marks)**
b) $2 \times 3^2 - (10 - 4)$ c) $(45 - 3^2) \div (4 + 2)$ d) $5^2 - \sqrt{64}$.

**Show your working**

**Answer:** _____

**2** **Task 2: Mechanics Braking Formula:** Stopping distance formula: **(3 marks)**
$d = 0.05v^2 + 0.2v$, where v = speed in mph. Find stopping distance at 60 mph.

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 11: BIDMAS & Braking Formula • Candidate Workspace

✎ Show all working for Worksheet 11 tasks below. Check your final answers.

DRILL 12 OF 34 • LEVEL 2 STANDARDS

# 12 Worksheet 12: Metric Conversions & Medicine

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Rapid Conversion Drills:** Convert: a) 4.8 kg into grams b) 650 ml into litres c) 2.4 m into cm d) 35 miles to km (5 mi ≈ 8 km). **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Hospital Patient Medication:** A patient requires 15 ml of oral antibiotic 3 times daily for 10 days. How many 200 ml bottles must be dispensed? **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 12: Metric Conversions & Medicine • Candidate Workspace

✎ Show all working for Worksheet 12 tasks below. Check your final answers.

DRILL 13 OF 34 • LEVEL 2 STANDARDS

13

# Worksheet 13: Elapsed Time & Meeting

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Elapsed Time Warm-Up:** Find elapsed duration: a) 08:45 to 14:15 b) 19:50 to 02:25 (next day). **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Train & Meeting Connection:** A business consultant catches the 09:28 train. Travel time is 1 hr 45 mins, followed by a 15-minute taxi ride. Does she arrive before 11:30? **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 13: Elapsed Time & Meeting • Candidate Workspace

✎ Show all working for Worksheet 13 tasks below. Check your final answers.

DRILL 14 OF 34 • LEVEL 2 STANDARDS

# 14 Worksheet 14: Overtime & Income Tax

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Overtime Rate Calculations:** Standard rate £12.40/hr. **(4 marks)**
Find: a) Time-and-a-half rate b) Double time rate c) Pay for 35 normal hrs + 6 hrs at time-and-a-half.

**Show your working**

**Answer:** _____

**2** **Task 2: Annual PAYE Income Tax:** A software developer earns £32,570 per annum. Personal allowance is £12,570. Basic rate tax is 20%. Calculate his annual income tax. **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 14: Overtime & Income Tax • Candidate Workspace

✎ Show all working for Worksheet 14 tasks below. Check your final answers.

DRILL 15 OF 34 • LEVEL 2 STANDARDS

# 15 Worksheet 15: Perimeter & Skirting Board

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: L-Shape Perimeter Warm-Up:** An L-shaped room has overall length 9 m and overall width 7 m. Cutout corner is 4 m by 3 m. Find its perimeter. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Skirting Board Cost:** A rectangular bedroom measures 4.8 m by 3.6 m. It has a 0.9 m door opening. Skirting board costs £4.50 per metre. Calculate cost. **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 15: Perimeter & Skirting Board • Candidate Workspace

✎ Show all working for Worksheet 15 tasks below. Check your final answers.

DRILL 16 OF 34 • LEVEL 2 STANDARDS

16

# Worksheet 16: Circumference & Semicircular Bed

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Circumference Drills:** Calculate circumference (take π = 3.14): a) Circle with diameter 15 cm b) Circle with radius 6.2 m. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Semicircular Flowerbed Border:** A garden has a semicircular flowerbed with diameter 8 m. Brick edging is placed around the entire perimeter. Calculate total length. (Use π = 3.142) **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 16: Circumference & Semicircular Bed • Candidate Workspace

✎ Show all working for Worksheet 16 tasks below. Check your final answers.

DRILL 17 OF 34 • LEVEL 2 STANDARDS

17

# Worksheet 17: Area & Laminate Flooring

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Area Formula Warm-Up:** Calculate area: a) Triangle: base 12 cm, height 7 cm b) Parallelogram: base 9.5 m, perpendicular height 4.2 m. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Laminate Flooring & 10% Wastage:** A dining room measures 5.4 m by 3.8 m. Flooring packs cover 2.2 m² each. Adding 10% for wastage, how many packs are needed? **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 17: Area & Laminate Flooring • Candidate Workspace

✎ Show all working for Worksheet 17 tasks below. Check your final answers.

DRILL 18 OF 34 • LEVEL 2 STANDARDS

# 18 Worksheet 18: Trapezium & Pond Paving

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Trapezium Area Warm-Up:** Calculate area: a) Trapezium: parallel sides 6 m and 10 m, height 4 m b) Circle of radius 5 m (take π = 3.14). **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Paving Around a Circular Pond:** A square lawn of side 12 m has a circular pond of diameter 4 m in its centre. What is the area of grass remaining? (Use π = 3.142) **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 18: Trapezium & Pond Paving • Candidate Workspace

✎ Show all working for Worksheet 18 tasks below. Check your final answers.

DRILL 19 OF 34 • LEVEL 2 STANDARDS

19

# Worksheet 19: Surface Area & Tent Fabric

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Cuboid Surface Area Warm-Up:** Calculate surface area: **(4 marks)**
a) Cuboid 10 cm × 6 cm × 4 cm b) Cube of side 5 cm.

**Show your working**

**Answer:** _____

**2** **Task 2: Triangular Prism Tent Fabric:** A tent is a triangular prism. Triangular ends have base 2.4 m, height 1.6 m, and slant edges 2.0 m. Length is 3.5 m. Find total fabric area. **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 19: Surface Area & Tent Fabric • Candidate Workspace

✎ Show all working for Worksheet 19 tasks below. Check your final answers.

DRILL 20 OF 34 • LEVEL 2 STANDARDS

20

# Worksheet 20: Volume & Water Cistern

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Volume Calculation Warm-Up:** Calculate volume: a) Cuboid 8 cm × 5 cm × 12 cm b) Cylinder: radius 4 cm, height 10 cm (take π = 3.14). **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Water Storage Cistern:** A rectangular water tank is 1.8 m long, 1.2 m wide, and 0.8 m deep. How many litres of water does it hold when full? **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 20: Volume & Water Cistern • Candidate Workspace

✎ Show all working for Worksheet 20 tasks below. Check your final answers.

DRILL 21 OF 34 • LEVEL 2 STANDARDS

21

# Worksheet 21: Prism Density & Swimming Pool

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Prism Volume Drills:** A prism has cross-sectional area 45 cm² and length 18 cm. What is its volume? If it is made of aluminium with density 2.7 g/cm³, find its mass. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Sloping Swimming Pool Volume:** A pool is 15 m long and 6 m wide. Depth slopes uniformly from 1.0 m to 2.0 m. Calculate total volume of water in cubic metres. **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 21: Prism Density & Swimming Pool • Candidate Workspace

✎ Show all working for Worksheet 21 tasks below. Check your final answers.

DRILL 22 OF 34 • LEVEL 2 STANDARDS

22

# Worksheet 22: Scale Drawings & Floor Plan

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Scale Factor Warm-Up:** On a plan with scale 1 : 50: a) A wall is 6.5 cm on plan. Real length in metres? b) A sofa is 1.8 m long in reality. Plan length in cm? **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Office Suite Floor Plan:** On a 1 : 75 scale drawing, an office measures 12 cm by 8 cm. Find its real ground floor area in square metres. **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 22: Scale Drawings & Floor Plan • Candidate Workspace

✎ Show all working for Worksheet 22 tasks below. Check your final answers.

DRILL 23 OF 34 • LEVEL 2 STANDARDS

# 23 Worksheet 23: Bearings & Ship Course

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Bearings Basics Warm-Up:** Write as 3-figure bearings: a) North-East b) South-West c) Due West d) 35° East of North. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Ship Course & Return Bearing:** A coastguard cutter sails from port on a bearing of 078° to reach a vessel in distress. What bearing must it steer to return to port? **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 23: Bearings & Ship Course • Candidate Workspace

✎ Show all working for Worksheet 23 tasks below. Check your final answers.

DRILL 24 OF 34 • LEVEL 2 STANDARDS

# 24 Worksheet 24: Speed, Density & Courier Route

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Formula Calculations Warm-Up:** a) Speed: 105 miles in 1 hr 45 mins. b) Density: mass 840 g, volume 120 cm³. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Courier Van Delivery Route:** A delivery van travels 90 miles on a motorway at an average speed of 60 mph, then 15 miles through city traffic at 20 mph. Find total time. **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 24: Speed, Density & Courier Route • Candidate Workspace

✎ Show all working for Worksheet 24 tasks below. Check your final answers.

DRILL 25 OF 34 • LEVEL 2 STANDARDS

25

# Worksheet 25: Two-Way Tables & Transport

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Complete the Table:** 50 office workers surveyed on transport mode: 28 are Female. 12 Males drive. 18 Females take public transport. 6 Males walk. No Females walk. Complete the table. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Probability from Table:** From the table in Task 1, what fraction of Male workers drive? Give your answer in simplest form. **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 25: Two-Way Tables & Transport • Candidate Workspace

✎ Show all working for Worksheet 25 tasks below. Check your final answers.

DRILL 26 OF 34 • LEVEL 2 STANDARDS

# 26 Worksheet 26: Dual Bar Charts & Sales Growth

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Read Dual Bar Chart:** A dual bar chart shows laptop sales (Units): Q1 (Store A: 120, Store B: 90); Q2 (Store A: 150, Store B: 180). Find total units sold and Store B's share. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Percentage Growth Comparison:** Store A sales grew from 120 to 150. Store B sales grew from 90 to 180. Calculate the percentage increase for both stores. Which grew faster? **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 26: Dual Bar Charts & Sales Growth • Candidate Workspace

✎ Show all working for Worksheet 26 tasks below. Check your final answers.

DRILL 27 OF 34 • LEVEL 2 STANDARDS

# 27 Worksheet 27: Line Graphs & Energy Trend

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Temperature Monitoring:** Patient temperature logged: 08:00 (38.6°C), 10:00 (38.2°C), 12:00 (37.4°C). Estimate temperature at 09:00. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Energy Consumption Trend:** Electricity meter reading: Mon 420 kWh, Tue 460 kWh, Wed 510 kWh, Thu 480 kWh, Fri 430 kWh. Find mean daily usage. **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 27: Line Graphs & Energy Trend • Candidate Workspace

✎ Show all working for Worksheet 27 tasks below. Check your final answers.

DRILL 28 OF 34 • LEVEL 2 STANDARDS

# 28 Worksheet 28: Pie Charts & Voter Angle

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Sector Angle Calculations:** 90 employees travel to work: Car (45), Bus (25), Train (15), Cycle (5). Calculate the sector angle for each travel mode. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Extracting Values from Angle:** In a pie chart of 1,200 voters, the sector for Candidate Green has an angle of 54°. How many people voted for Candidate Green? **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 28: Pie Charts & Voter Angle • Candidate Workspace

✎ Show all working for Worksheet 28 tasks below. Check your final answers.

DRILL 29 OF 34 • LEVEL 2 STANDARDS

# 29 Worksheet 29: Averages & Adding Values

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Calculate the 4 Measures:** For data set: 14, 18, 12, 25, 18, 16, 23. Find: a) Mean b) Median c) Mode d) Range. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Adding a New Value to Mean:** A call centre agent has a mean call time of 4.5 minutes across 8 calls. Her 9th call lasts 9.0 minutes. What is her new mean? **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 29: Averages & Adding Values • Candidate Workspace

✎ Show all working for Worksheet 29 tasks below. Check your final answers.

DRILL 30 OF 34 • LEVEL 2 STANDARDS

# 30 Worksheet 30: Grouped Frequency & Estimated Mean

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Midpoints & Products Warm-Up:** Classes: 0-10 (freq 4), 10-20 (freq 11), 20-30 (freq 5). State midpoints and calculate Sum(fx). **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Estimated Mean Calculation:** Using Task 1 above, calculate the estimated mean. Why is this an estimate? **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 30: Grouped Frequency & Estimated Mean • Candidate Workspace

✎ Show all working for Worksheet 30 tasks below. Check your final answers.

DRILL 31 OF 34 • LEVEL 2 STANDARDS

31

# Worksheet 31: Data Comparison & Drivers

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Delivery Driver Performance:** Driver X: Mean deliveries = 32, Range = 4. Driver Y: Mean deliveries = 36, Range = 15. Compare their performance. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Hospital Waiting Times:** Hospital A median wait = 42 mins, IQR = 15 mins. Hospital B median wait = 35 mins, IQR = 28 mins. Advise a patient. **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 31: Data Comparison & Drivers • Candidate Workspace

✎ Show all working for Worksheet 31 tasks below. Check your final answers.

DRILL 32 OF 34 • LEVEL 2 STANDARDS

32

# Worksheet 32: Scatter Graphs & Line of Best Fit

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Correlation Identification:** State correlation: a) Revision hours and exam score b) Car engine size and mpg c) Shoe size and salary. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Reading Line of Best Fit:** A scatter graph relates temperature (°C) to cold drinks sold. The line passes through (15, 60) and (25, 100). Predict drinks at 20°C. **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 32: Scatter Graphs & Line of Best Fit • Candidate Workspace

✎ Show all working for Worksheet 32 tasks below. Check your final answers.

DRILL 33 OF 34 • LEVEL 2 STANDARDS

# 33 Worksheet 33: Probability Drills & Coin/Dice

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Basic Probability Drills:** A box has 8 red, 5 green, 7 blue counters. A counter is chosen at random: a) P(Red) b) P(Not Green) c) P(Yellow). **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Independent Combined Probability:** A fair coin is flipped and a standard 6-sided dice is rolled. What is the probability of getting a Head and a 6? **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 33: Probability Drills & Coin/Dice • Candidate Workspace

✎ Show all working for Worksheet 33 tasks below. Check your final answers.

DRILL 34 OF 34 • LEVEL 2 STANDARDS

# 34 Worksheet 34: Combined Probability & Weather

### ★ Guided Calculation Scaffold

**Step 1:** Identify the statutory formula and extract key numerical variables from the text.
**Step 2:** Substitute values into calculation and record intermediate units.
**Step 3:** Perform secondary check using reverse arithmetic or approximation.

**1** **Task 1: Complete Tree Branches:** Candidate sits two tests: P(Pass 1st) = 0.8. If pass, P(Pass 2nd) = 0.7. If fail, P(Pass 2nd) = 0.4. Find P(Pass both). **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2: Weather & Commute Delay:** P(Rain) = 0.3. If rain, P(Late) = 0.6. If dry, P(Late) = 0.1. Calculate total probability of being late. **(3 marks)**

**Show your working**

**Answer:** _____

5MM RULED GRID WORKING AREA

# Worksheet 34: Combined Probability & Weather • Candidate Workspace

✎ Show all working for Worksheet 34 tasks below. Check your final answers.

CP1

CANDIDATE ASSESSMENT FRAME

# Checkpoint Assessment 1 • Part 1/4

**1** **Question 1:** Calculate without using a calculator: 416 × 28. Show your full column method. **(2 marks)**

**Show your working**

**Answer:** _____

**2** **Question 2:** A freezer compartment is at -14°C. The thermostat is adjusted so the temperature rises by 6.5°C, then drops by 3.2°C. What is the new temperature? **(2 marks)**

**Show your working**

**Answer:** _____

**3** **Question 3:** Evaluate giving your answer as a mixed number in simplest form: 1 3/5 + 2 1/4. **(2 marks)**

**Show your working**

**Answer:** _____

CP1

CANDIDATE ASSESSMENT FRAME

# Checkpoint Assessment 1 • Part 2/4

**3** **Question 4:** Work out: 3,744 ÷ 13. Show your full written division method. **(2 marks)**

**Show your working**

**Answer:** _____

**4** **Question 5:** A tradesperson quotes £420.00 excluding VAT. VAT is charged at 20%. How much is the total quote with VAT included? **(2 marks)**

**Show your working**

**Answer:** _____

**5** **Question 6:** Share £1,260 between three apprentices in the ratio 3 : 2 : 1. How much does the apprentice with the smallest share receive? **(2 marks)**

**Show your working**

**Answer:** _____

CP1

CANDIDATE ASSESSMENT FRAME

# Checkpoint Assessment 1 • Part 3/4

**5** **Question 7:** A sports retailer sells running shoes for £68.00 after a 15% discount. Calculate the original full price before the discount was applied. **(3 marks)**

**Show your working**

**Answer:** _____

**6** **Question 8:** A company deposits £15,000 in a business growth account offering 3.8% compound interest per year for 3 years. Calculate the total balance in the account at the end of the 3 years. **(3 marks)**

**Show your working**

**Answer:** _____

**7** **Question 9:** It takes 4 assembly technicians 15 hours to assemble a batch of medical ventilators. The hospital requests the batch in 10 hours. How many technicians in total are required? **(3 marks)**

**Show your working**

**Answer:** _____

CP1

CANDIDATE ASSESSMENT FRAME

# Checkpoint Assessment 1 • Part 4/4

**7** **Question 10:** A concrete mix requires sand, ballast, and cement in the ratio 5 : 3 : 2. A builder prepares 3.6 tonnes of mix. He has 1.7 tonnes of sand in stock. Does he have enough sand? Show full working. **(3 marks)**

**Show your working**

**Answer:** _____

**8** **Question 11:** A logistics manager bought an electric delivery van for £32,000. It depreciates by 16% in the first year and 12% in the second year. What is the value of the van at the end of year 2? **(3 marks)**

**Show your working**

**Answer:** _____

**9** **Question 12:** An office manager checks an invoice: 12 desks at £145.50 each, 24 chairs at £68.00 each. A delivery charge of £45.00 is added. A bulk discount of 7.5% is deducted from the total before adding 20% VAT. What is the final invoice total? **(3 marks)**

**Show your working**

**Answer:** _____

CANDIDATE ASSESSMENT FRAME

CP2

# Checkpoint Assessment 2 • Part 1/4

**1** **Question 1:** A delivery truck travels 135 miles in 2 hours 15 minutes. Calculate its average speed in miles per hour. **(2 marks)**

**Show your working**

**Answer:** _____

**2** **Question 2:** Convert: a) 4.2 kg to grams b) 650 ml to litres c) 25 miles to km using 5 miles = 8 km. **(2 marks)**

**Show your working**

**Answer:** _____

**3** **Question 3:** A circle has a diameter of 14 cm. Using π = 22/7, calculate: a) its circumference b) its area. **(2 marks)**

**Show your working**

**Answer:** _____

CP2

CANDIDATE ASSESSMENT FRAME

# Checkpoint Assessment 2 • Part 2/4

**3** **Question 4:** A rectangular courtyard measures 12 m by 8.5 m. In the centre is a square fountain with sides of 2.5 m. Calculate the paved area of the courtyard. **(2 marks)**

**Show your working**

**Answer:** _____

**4** **Question 5:** A clock shows 09:35. An online training conference lasts for 3 hours 48 minutes. At what time does the conference end? (Use 24-hour clock format) **(2 marks)**

**Show your working**

**Answer:** _____

**5** **Question 6:** A warehouse worker earns £12.50 per hour for 36 hours. Overtime is paid at time-and-a-half. In one week he works 4 hours overtime. What is his gross pay? **(2 marks)**

**Show your working**

**Answer:** _____

CP2

CANDIDATE ASSESSMENT FRAME

# Checkpoint Assessment 2 • Part 3/4

**5** **Question 7:** A commercial grain silo is shaped as a cylinder with diameter 4.2 m and height 8.5 m. Calculate the volume of the silo in cubic metres to 1 decimal place. (Use π = 3.142) **(3 marks)**

**Show your working**

**Answer:** _____

**6** **Question 8:** A metal casting has a mass of 3,570 g and a volume of 420 cm³. Calculate its density in g/cm³. If pure zinc has a density of 7.14 g/cm³, determine whether the casting is pure zinc. **(3 marks)**

**Show your working**

**Answer:** _____

**7** **Question 9:** On a map with a scale of 1 : 25,000, the distance between two telecommunication masts is 16.4 cm. Calculate the actual distance between the masts in kilometres. **(3 marks)**

**Show your working**

**Answer:** _____

CP2

CANDIDATE ASSESSMENT FRAME

# Checkpoint Assessment 2 • Part 4/4

**7** **Question 10:** An office worker earns £34,000 gross per year. The tax-free personal allowance is £12,570. Basic rate income tax is charged at 20%. Employee National Insurance is 8% on earnings above £12,570. Calculate her total monthly take-home pay after tax and NI. **(3 marks)**

**Show your working**

**Answer:** _____

**8** **Question 11:** A lawn is in the shape of a trapezium with parallel sides 14 m and 22 m, and perpendicular distance 12 m. Grass seed costs £6.40 per box. Each box covers 25 m². How much does it cost to seed the entire lawn? **(3 marks)**

**Show your working**

**Answer:** _____

**9** **Question 12:** A heating engineer installs a closed cylindrical thermal expansion vessel: radius 0.35 m and height 1.2 m. The entire exterior surface is insulated with thermal foil costing £14.50/m². Calculate the cost of the foil. (Use π = 3.142) **(3 marks)**

**Show your working**

**Answer:** _____

CP3

CANDIDATE ASSESSMENT FRAME

# Checkpoint Assessment 3 • Part 1/4

**1** **Question 1:** The table shows the number of patients seen at a dental practice: Mon 24, Tue 32, Wed 28, Thu 36, Fri 30. Calculate: a) the mean b) the range. **(2 marks)**

**Show your working**

**Answer:** _____

**2** **Question 2:** A spinner has 8 equal sections numbered 1 to 8. Calculate the probability of spinning: a) an even number b) a multiple of 3. **(2 marks)**

**Show your working**

**Answer:** _____

**3** **Question 3:** 60 office workers were surveyed on their preferred lunch option: Sandwiches (30), Hot Food (20), Salad (10). Calculate the pie chart sector angle for each option. **(2 marks)**

**Show your working**

**Answer:** _____

CANDIDATE ASSESSMENT FRAME

CP3

# Checkpoint Assessment 3 • Part 2/4

**3** **Question 4:** A box contains 5 red pens, 7 blue pens, and 8 black pens. A pen is picked at random. What is the probability that it is NOT red? Give your answer as a fraction in simplest form. **(2 marks)**

**Show your working**

**Answer:** _____

**4** **Question 5:** The test marks of 9 apprentices are: 45, 52, 58, 61, 64, 68, 72, 75, 85. Find: a) the median mark b) the interquartile range. **(2 marks)**

**Show your working**

**Answer:** _____

**5** **Question 6:** State whether each pair of variables has positive, negative, or no correlation: a) Daily sunshine hours and solar panel output b) Car engine mileage and market resale value. **(2 marks)**

**Show your working**

**Answer:** _____

CP3

CANDIDATE ASSESSMENT FRAME

# Checkpoint Assessment 3 • Part 3/4

**5** **Question 7:** A delivery depot records the delivery times (t mins) for 50 consignments: $0 < t \leq 10$ (12), $10 < t \leq 20$ (22), $20 < t \leq 30$ (11), $30 < t \leq 40$ (5). Calculate the estimated mean delivery time. **(3 marks)**

**Show your working**

**Answer:** _____

**6** **Question 8:** A quality assurance inspector tests computer microchips. Machine A produces 60% of chips with a 2% defect rate. Machine B produces 40% of chips with a 5% defect rate. What is the overall probability that a randomly chosen chip is defective? **(3 marks)**

**Show your working**

**Answer:** _____

**7** **Question 9:** A scatter graph plots vehicle weight (kg) against fuel economy (mpg). The line of best fit passes through (1200, 48) and (1800, 32). Predict the fuel economy for a vehicle weighing 1500 kg, and describe the relationship. **(3 marks)**

**Show your working**

**Answer:** _____

CANDIDATE ASSESSMENT FRAME

CP3

# Checkpoint Assessment 3 • Part 4/4

**7** **Question 10:** A supermarket manager audits 200 customer transactions. 120 customers paid by contactless card, 50 by mobile app, and 30 in cash. Of the contactless users, 40% spent over £30. Of the mobile app users, 60% spent over £30. Construct a two-way table and find the probability that a customer picked at random spent over £30. **(3 marks)**

**Show your working**

**Answer:** _____

**8** **Question 11:** Two recruitment consultants interview candidates. Consultant X rates 8 candidates with a mean score of 72. Consultant Y rates 12 candidates with a mean score of 65. Calculate the combined mean score for all 20 candidates. **(3 marks)**

**Show your working**

**Answer:** _____

**9** **Question 12:** The probability of train service A being delayed is 0.20. The probability of train service B being delayed is 0.15. Assuming delays are independent, calculate the probability that at least one train is on time. **(3 marks)**

**Show your working**

**Answer:** _____

CS1

SYNOPTIC REAL-WORLD PROBLEM

# Case Study 1: NHS Facilities & Ward Refurbishment

### Project Specification

Calculate flooring area, antimicrobial paint requirements, and project budget compliance.

Use the data provided to complete the project costing and compliance audit.

**1** **Task 1 (Flooring & Paint Costing):** A rectangular hospital ward measures 15 m long by 8 m wide, with walls 3 m high. Anti-slip vinyl flooring costs £24.50 per m². Antimicrobial wall paint covers 12 m² per tin and each tin costs £38.00 (ignore doors/windows for safety margin). Calculate the combined material cost for the floor and 2 coats of wall paint. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2 (Labour & Budget Compliance):** Three Estates operatives working 8 hours per day can complete the refurbishment in 5 days at £18.50 per hour each. If the Ward Manager has a total refurbishment budget of £5,200 (materials + labour), determine whether the project is within budget and state the surplus or deficit. **(4 marks)**

**Show your working**

**Answer:** _____

PROJECT CALCULATIONS FRAME

# Case Study 1 • Candidate Workspace

✎ **Candidate full calculation working and decision summary:**

SYNOPTIC REAL-WORLD PROBLEM

CS2

# Case Study 2: Logistics Hub & Pallet Loading Optimization

## Project Specification

Determine volume capacities, weight axle tolerances, and delivery fuel efficiency.

Use the data provided to complete the project costing and compliance audit.

**1** **Task 1 (Container Volume & Payload Limit):** An articulated trailer has internal dimensions 13.2 m long, 2.4 m wide, and 2.6 m high. Standard Euro-pallets measure 1.2 m by 0.8 m and are stacked to a height of 1.3 m (allowing 2 tiers high). Each loaded pallet weighs 380 kg. Calculate the maximum number of pallets that fit in the trailer and check if their total weight is within the 26-tonne axle payload limit. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2 (Fleet Fuel & Delivery Schedule):** The loaded lorry travels 210 miles at an average speed of 50 mph, plus a 45-minute statutory tachograph break. Fuel consumption is 8.4 miles per gallon (1 gallon = 4.55 litres) and diesel costs £1.48 per litre. Calculate the total journey time and total fuel cost. **(4 marks)**

**Show your working**

**Answer:** _____

PROJECT CALCULATIONS FRAME

# Case Study 2 • Candidate Workspace

✎ **Candidate full calculation working and decision summary:**

SYNOPTIC REAL-WORLD PROBLEM

CS3

# Case Study 3: Commercial Construction Site & Concrete Pour

### Project Specification

Calculate trench volumes, foundation mix ratios, and site labour scheduling.

Use the data provided to complete the project costing and compliance audit.

**1** **Task 1 (Trench Volume & Mix Ratio):** A foundation trench is 25 m long, 0.8 m wide, and 1.2 m deep. Concrete has a density of 2,400 kg/m³ and is mixed using cement, sand, and aggregate in the ratio 1 : 2 : 3 by mass. Calculate the trench volume in m³ and the mass of cement required in tonnes. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Task 2 (Site Labour Inverse Proportion & Cost):** Six groundworkers take 8 working days to excavate and pour the foundations. To beat forecasted frost, the site manager wants the work completed in 4 working days. How many additional groundworkers must be hired, and what is the total labour bill if each worker is paid £165 per day? **(4 marks)**

**Show your working**

**Answer:** _____

PROJECT CALCULATIONS FRAME

# Case Study 3 • Candidate Workspace

✎ Candidate full calculation working and decision summary:

P1

OFFICIAL STYLE EXAMINATION BOOKLET

# Practice Paper 1 • Page 1 of 8

**1** **Question 1:** Calculate without using a calculator: 648 × 35. Show your full written method. **(3 marks)**

**Show your working**

**Answer:** _____

**2** **Question 2:** Evaluate giving your answer as a mixed number in simplest form: 2 2/3 ÷ 4/9. **(2 marks)**

**Show your working**

**Answer:** _____

P1

OFFICIAL STYLE EXAMINATION BOOKLET

# Practice Paper 1 • Page 2 of 8

**3** **Question 3:** A kitchen fitter cuts a circular hole with diameter 28 cm in a wooden worktop. Using π = 22/7, calculate the perimeter (circumference) of the cut hole. **(3 marks)**

**Show your working**

**Answer:** _____

**4** **Question 4:** An invoice shows a power saw sold for £264.00 including 20% VAT. Calculate the cost of the saw before VAT was added. **(3 marks)**

**Show your working**

**Answer:** _____

P1

OFFICIAL STYLE EXAMINATION BOOKLET

# Practice Paper 1 • Page 3 of 8

**5** **Question 5:** A healthcare assistant prepares 750 ml of saline infusion. Each dose is 60 ml. How many full doses can be administered, and how many millilitres remain? **(2 marks)**

**Show your working**

**Answer:** _____

**6** **Question 6:** Share £2,400 between three apprentices in the ratio 4 : 3 : 1. What is the difference between the largest share and the smallest share? **(3 marks)**

**Show your working**

**Answer:** _____

P1

OFFICIAL STYLE EXAMINATION BOOKLET

# Practice Paper 1 • Page 4 of 8

**7** **Question 7:** A senior nurse has a gross salary of £36,570. The personal allowance is £12,570. Basic rate income tax is 20%. National Insurance is 8% of earnings above £12,570. A workplace pension deduction of 6% of gross salary is made. Calculate her net monthly take-home pay. **(4 marks)**

**Show your working**

**Answer:** _____

**8** **Question 8:** An engineering firm purchases a computerized CNC milling machine for £45,000. The machine depreciates at a compound rate of 15% per annum. Calculate the book value of the machine after 3 years. **(4 marks)**

**Show your working**

**Answer:** _____

P1

OFFICIAL STYLE EXAMINATION BOOKLET

# Practice Paper 1 • Page 5 of 8

**9** **Question 9:** A heavy haulage transporter travels 168 miles. The driver leaves depot at 06:15 and takes a mandatory 45-minute rest break during the journey. The vehicle arrives at 10:30. Calculate the driver's average driving speed in mph while moving. **(4 marks)**

**Show your working**

**Answer:** _____

**10** **Question 10:** A landscape contractor constructs a patio in the shape of a trapezium with parallel sides 8.5 m and 13.5 m, and perpendicular height 6.4 m. Paving slabs cost £28.50 per square metre. Sand and mortar bedding adds £4.50 per square metre. Calculate the total material cost. **(4 marks)**

**Show your working**

**Answer:** _____

P1

OFFICIAL STYLE EXAMINATION BOOKLET

# Practice Paper 1 • Page 6 of 8

**11** **Question 11:** A commercial stainless-steel cylindrical storage tank has an internal diameter of 2.4 m and a height of 3.5 m. Calculate the liquid capacity of the tank in litres. (Use π = 3.142) **(4 marks)**

**Show your working**

**Answer:** _____

**12** **Question 12:** A local council tracks response times (hours) for emergency housing repairs: $0 < h \leq 12$ (18 calls), $12 < h \leq 24$ (32 calls), $24 < h \leq 36$ (14 calls), $36 < h \leq 48$ (6 calls). Calculate the estimated mean response time in hours. **(4 marks)**

**Show your working**

**Answer:** _____

P1

OFFICIAL STYLE EXAMINATION BOOKLET

# Practice Paper 1 • Page 7 of 8

**13** **Question 13:** An architect designs an executive office suite: 12 m long by 8.5 m wide. A circular glass conference booth of diameter 4 m is located in the corner. The remaining floor space is carpeted with luxury tiles costing £32.00/m². VAT at 20% is added to the carpet cost. Calculate the total cost to carpet the office suite. (Use π = 3.142) **(5 marks)**

**Show your working**

**Answer:** _____

**14** **Question 14:** On an ordnance survey map drawn to a scale of 1 : 25,000, a wildlife reserve boundary is a rectangle measuring 14.8 cm by 9.2 cm. Calculate the real perimeter of the reserve in kilometres, and the real land area of the reserve in square kilometres. **(5 marks)**

**Show your working**

**Answer:** _____

P1

OFFICIAL STYLE EXAMINATION BOOKLET

# Practice Paper 1 • Page 8 of 8

**15** **Question 15:** A healthcare audit tracks 150 clinic patients tested for diabetes: 90 patients are over 50 years old. Of those over 50, 27 tested positive. Of those aged 50 or under, 12 tested positive. Construct a complete two-way table. A patient is chosen at random from those who tested positive. What is the probability that this patient is aged 50 or under? **(5 marks)**

**Show your working**

**Answer:** _____

**16** **Question 16:** A manufacturing plant produces precision steel rods. A sample of 100 rods has a mean length of 148.5 mm with a range of 4.2 mm. A second production line produces 100 rods with a mean length of 148.4 mm with a range of 1.1 mm. The target length is 148.5 mm. Compare the performance of the two production lines and advise the plant manager which line produces higher quality output. **(5 marks)**

**Show your working**

**Answer:** _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

# Practice Paper 2 • Page 1 of 8

**1** **Question 1:** Calculate: 684 ÷ 18 without using a calculator. Show your full working. **(4 marks)**

**Show your working**

**Answer:** _____

**2** **Question 2:** A pallet measures 1.2 m by 0.8 m. A warehouse floor is 24 m by 16 m. Calculate the maximum number of pallets that can fit on the floor in a single layer. **(4 marks)**

**Show your working**

**Answer:** _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

# Practice Paper 2 • Page 2 of 8

**3** **Question 3:** Calculate: 3/5 × 4/7. Express your answer in simplest form. **(4 marks)**

**Show your working**

**Answer:** _____

**4** **Question 4:** A delivery driver's hourly rate increased from £12.50 to £13.75. Calculate the percentage pay increase. Check your answer. **(4 marks)**

**Show your working**

**Answer:** _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

# Practice Paper 2 • Page 3 of 8

**5** **Question 1:** Calculate: 345.8 − 187.94 without a calculator. **(4 marks)**
Show columnar working.

**Show your working**

**Answer:** _____

**6** **Question 2:** A retailer buys stock for £450 and sells it for £630. **(4 marks)**
Calculate the percentage profit margin on cost.

**Show your working**

**Answer:** _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

# Practice Paper 2 • Page 4 of 8

**7** **Question 5: Fleet Fuel Efficiency:** Van A travels 420 miles on 60 litres of fuel. Van B travels 385 miles on 52 litres. Which van achieves better miles-per-gallon (mpg)? (1 gallon = 4.55 litres). **(5 marks)**

**Show your working**

**Answer:** _____

**8** **Question 6: Cargo Weight & Density:** A container has volume 36 $m^3$. It is loaded with cargo having an average density of 650 $kg/m^3$. The maximum payload limit is 24 tonnes. Is the container overloaded? **(5 marks)**

**Show your working**

**Answer:** _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

# Practice Paper 2 • Page 5 of 8

**9** **Question 7: Courier Delivery Routes:** Driver covers 168 miles in 3 hours 30 mins. A 45-minute delivery stop was included. What was the driving speed while moving? **(5 marks)**

**Show your working**

**Answer:** _____

**10** **Question 8: Forklift Ramp Slope:** A loading ramp rises 1.2 m over a horizontal distance of 5 m. Calculate the length of the sloping surface using Pythagoras ($a^2 + b^2 = c^2$). **(5 marks)**

**Show your working**

**Answer:** _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

# Practice Paper 2 • Page 6 of 8

**11** **Question 9: Delivery Time Statistics:** Delivery times (mins): 10-19 (5), 20-29 (14), 30-39 (21), 40-49 (10). Calculate the estimated mean delivery time. **(5 marks)**

**Show your working**

**Answer:** _____

**12** **Question 10: Warehouse Racking:** Racking is offered with a 12% discount. The discounted cost is £3,960. Find the original price before discount. **(5 marks)**

**Show your working**

**Answer:** _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

# Practice Paper 2 • Page 7 of 8

**13** **Question 11: Currency & Import:** A consignment from Germany costs €14,500. Customs import duty is 6%. If £1 = €1.15, calculate the total cost in GBP including duty. **(5 marks)**

**Show your working**

**Answer:** _____

**14** **Question 12: Delivery Success Rate:** Out of 850 deliveries, 34 were failed deliveries. Calculate the percentage success rate to 1 decimal place. **(5 marks)**

**Show your working**

**Answer:** _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

# Practice Paper 2 • Page 8 of 8

**15** **Question 13: Storage Space Volume:** A cold storage room is 12m long, 8m wide, and 3.5m high. Industrial cooling requires 45 Watts per $m^3$. What power cooling unit is required? **(4 marks)**

**Show your working**

**Answer:** _____

**16** **Question 14: Van Tyre Replacement:** Tyres cost £95 each. A fleet replaces all 4 tyres on 15 vans. The supplier offers 'Buy 3 get 1 free' on every van. Calculate total cost. **(4 marks)**

**Show your working**

**Answer:** _____

MODEL ANSWERS & MARKING CRITERIA

# Answer Key • Part 1 of 3

## ▶ Worksheets 1–12 Step-by-Step Model Answers

**W1: T1:** a) -23; b) 30; c) 42; d) -9. **T2:** Closing balance = -2340 + 5800 - 1950 - 780 - 1150 = **-£420.00** (in overdraft).

**W2: T1:** a) 348 × 27 = **9,396**; b) 516 × 38 = **19,608**. **T2:** 918 ÷ 24 = **38.25**.

**W3: T1:** a) (10+9) ÷ 24 = **19/24**; b) 9/4 - 5/3 = (27-20) ÷ 12 = **7/12**. **T2:** Cut = 1 3/4 + 2 3/8 = 4 1/8 m. Remaining = 5 - 4 1/8 = **7/8 m** (0.875 m).

**W4: T1:** a) 12/72 = **1/6**; b) 7/2 × 10/7 = **5**. **T2:** Batches = 18 ÷ 2 1/4 = 18 × 4/9 = **8 batches**.

**W5: T1:** a) 34.232; b) 62.36; c) 1.61; d) 31.5. **T2:** Pre-VAT = (4 × 14.25) + 68.50 + (2.5 × 42) = 57 + 68.50 + 105 = **£230.50**.

**W6: T1:** a) 1.175; b) 0.88; c) 1.045; d) 0.0825. **T2:** Discount = £650 × 0.22 = £143. Sale price = £650 × 0.78 = **£507.00**.

**W7: T1:** a) 144 ÷ 1.20 = **£120.00**; b) 204 ÷ 0.85 = **£240.00**. **T2:** Pre-VAT = £348 ÷ 1.20 = **£290.00**.

**W8: T1:** a) 560 ÷ 7 = 80 → **£400 and £160**; b) 15 cm : 120 cm = **1 : 8**. **T2:** Difference = 3 parts = £21,000 → 1 part = £7,000. Total profit (11 parts) = **£77,000**.

**W9: T1:** a) 4 × 6 = 24 → 24 ÷ 3 = **8 days**; b) 240 ÷ 48 = **5 hours**. **T2:** Brand X = 28.50 / 5 = £5.70/L; Brand Y = 41.25 / 7.5 = £5.50/L. **Brand Y is better value**.

**W10: T1:** a) 4000 × $(1.03)^2$ = **£4,243.60**; b) 12000 × $(0.9)^3$ = **£8,748.00**. **T2:** £24,000 × 0.85 × 0.88 = **£17,952.00**.

**W11: T1:** a) 11; b) 12; c) 6; d) 17. **T2:** d = 0.05($60^2$) + 0.2(60) = 0.05(3600) + 12 = 180 + 12 = **192 feet**.

**W12: T1:** a) 4,800 g; b) 0.65 L; c) 240 cm; d) 56 km. **T2:** Total dose = 15 × 3 × 10 = 450 ml. Bottles needed = 450 ÷ 200 = 2.25 → **3 bottles**.

## ✎ Personal Learning Reflections (Part 1)

MODEL ANSWERS & MARKING CRITERIA

# Answer Key • Part 2 of 3

## ▶ Worksheets 13–24 Step-by-Step Model Answers

**W13: T1:** a) 5h 30m; b) 6h 35m. **T2:** 09:28 + 1h 45m = 11:13 + 15m = 11:28. **Yes, arrives before 11:30** (at 11:28).

**W14: T1:** a) £18.60; b) £24.80; c) (35 × 12.40) + (6 × 18.60) = 434 + 111.60 = **£545.60**. **T2:** Taxable = £20,000. Tax (20%) = **£4,000.00**.

**W15: T1:** P = 2(9 + 7) = **32 m**. **T2:** Perimeter = 2(4.8 + 3.6) - 0.9 = 16.8 - 0.9 = 15.9 m. Cost = 15.9 × 4.50 = **£71.55**.

**W16: T1:** a) 3.14 × 15 = **47.1 cm**; b) 2 × 3.14 × 6.2 = **38.94 m**. **T2:** Curved = 1/2 π d = 1/2(3.142)(8) = 12.57 m. Total P = 12.57 + 8 = **20.57 m**.

**W17: T1:** a) 1/2(12)(7) = **42 cm²**; b) 9.5 × 4.2 = **39.9 m²**. **T2:** Area = 5.4 × 3.8 = 20.52 m². With 10% = 22.572 m². Packs = 22.572 ÷ 2.2 = 10.26 → **11 packs**.

**W18: T1:** a) (6+10) ÷ 2 × 4 = **32 m²**; b) 3.14 × 25 = **78.5 m²**. **T2:** Lawn = 144 m². Pond = 3.142 × 4 = 12.57 m². Remaining grass = **131.43 m²**.

**W19: T1:** a) 2(60 + 40 + 24) = **248 cm²**; b) 6 × 25 = **150 cm²**. **T2:** Ends = 2(1/2 × 2.4 × 1.6) = 3.84 m². Sides = 2(2.0 × 3.5) = 14 m². Base = 2.4 × 3.5 = 8.4 m². Total = **26.24 m²**.

**W20: T1:** a) 8 × 5 × 12 = **480 cm³**; b) 3.14 × 16 × 10 = **502.4 cm³**. **T2:** V = 1.8 × 1.2 × 0.8 = 1.728 m³ = **1,728 litres**.

**W21: T1:** V = 45 × 18 = 810 cm³. Mass = 810 × 2.7 = **2,187 g** (2.187 kg). **T2:** Mean depth = 1.5 m. V = 15 × 6 × 1.5 = **135 m³**.

**W22: T1:** a) 6.5 × 50 = 325 cm = **3.25 m**; b) 180 ÷ 50 = **3.6 cm**. **T2:** Real l = 12 × 0.75 = 9 m; w = 8 × 0.75 = 6 m. Area = 9 × 6 = **54 m²**.

**W23: T1:** a) 045°; b) 225°; c) 270°; d) 035°. **T2:** Return bearing = 078° + 180° = **258°**.

**W24: T1:** a) 105 ÷ 1.75 = **60 mph**; b) 840 ÷ 120 = **7.0 g/cm³**. **T2:** Motorway = 90/60 = 1.5 hrs; City = 15/20 = 0.75 hrs. Total = 2.25 hrs = **2 hours 15 minutes**.

## ✎ Personal Learning Reflections (Part 2)

MODEL ANSWERS & MARKING CRITERIA

# Answer Key • Part 3 of 3

## Worksheets 25–34 & Case Studies 1–3 Model Answers

**W25: T1:** Male: Drive 12, Public 4, Walk 6, Total 22. Female: Drive 10, Public 18, Walk 0, Total 28. **T2:** Fraction of males driving = 12/22 = **6/11**.

**W26: T1:** Total units = (120+90) + (150+180) = 540. Store B share = 270/540 = **50%**. **T2:** Store A = (30/120) × 100 = 25%; Store B = (90/90) × 100 = 100%. **Store B grew faster**.

**W27: T1:** Estimate at 09:00: mid-point between 38.6 and 38.2 = **38.4°C**. **T2:** Mean daily usage = (420+460+510+480+430) ÷ 5 = 2300 ÷ 5 = **460 kWh**.

**W28: T1:** Car = 180°, Bus = 100°, Train = 60°, Cycle = 20°. **T2:** Votes = (54/360) × 1,200 = **180 votes**.

**W29: T1:** a) Mean = 126/7 = **18**; b) Median = **18**; c) Mode = **18**; d) Range = 25-12 = **13**. **T2:** Total sum = (8 × 4.5) + 9.0 = 36 + 9 = 45. New mean = 45 ÷ 9 = **5.0 minutes**.

**W30: T1:** Midpoints: 5, 15, 25. Products: 20, 165, 125. Sum(fx) = 310. **T2:** Estimated mean = 310 ÷ 20 = **15.5**. It is an estimate because exact values within intervals are unknown.

**W31: T1:** Driver Y has higher mean deliveries (36 vs 32), but Driver X is much more consistent with a smaller range (4 vs 15). **T2:** Hospital B has lower median wait (35 vs 42 mins), but Hospital A has more consistent waits (IQR 15 vs 28 mins).

**W32: T1:** a) Positive; b) Negative; c) Zero. **T2:** At 20°C, drinks sold = **80 units** (midpoint between 60 and 100).

**W33: T1:** Total = 20. a) P(Red) = 8/20 = **2/5**; b) P(Not Green) = 15/20 = **3/4**; c) P(Yellow) = **0**. **T2:** P(Head and 6) = 1/2 × 1/6 = **1/12**.

**W34: T1:** P(Pass both) = 0.8 × 0.7 = **0.56** (56%). **T2:** P(Rain and Late) = 0.3 × 0.6 = 0.18. P(Dry and Late) = 0.7 × 0.1 = 0.07. Total P(Late) = 0.18 + 0.07 = **0.25** (25%).

**CS1 (NHS Ward):** Floor 120 m² (£2,940) + Paint 276 m² ÷ 12 = 23 tins (£874) = **£3,814 materials**. Labour 120 hrs × £18.50 = £2,220. Total = **£6,034 (£834 over budget)**.

**CS2 (Logistics Hub):** 11 × 3 × 2 = **66 pallets**; Mass = **25.08 t ≤ 26 t (OK)**. Time = **4h 57m**; Fuel = 25 gal = 113.75 L × £1.48 = **£168.35**.

**CS3 (Concrete Pour):** Volume = **24 m³**; Mass = 57.6 t → Cement (1/6) = **9.6 tonnes**. Workers for 4 days = 12 (**6 extra workers**); Labour = **£7,920.00**.

## ✎ Personal Learning Reflections (Part 3)

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